

HOW TO SET UP A PYTHON ENVIRONMENT FOR DATA SCIENCE & MACHINE LEARNING

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Why do you think learning python programming is important for civil engineers?



1. Data is transforming civil engineering:

- traffic data, bridge sensors data, cracks images data and material test results and structural vibration signals
- Python helps to handle and analyze these data more efficiently.

2. Python replaces repetitive tasks and manual computations.

3. Python works with modern engineering tools

- Like AutoCAD, GIS, BIM, Finite Element tools, sensors and SHM systems

4. Evolving of ML & AI in civil engineering

- Damage detection, predictive maintenance, soil classification, pavement deterioration forecasting

5. Python is open-source, free and have large community.

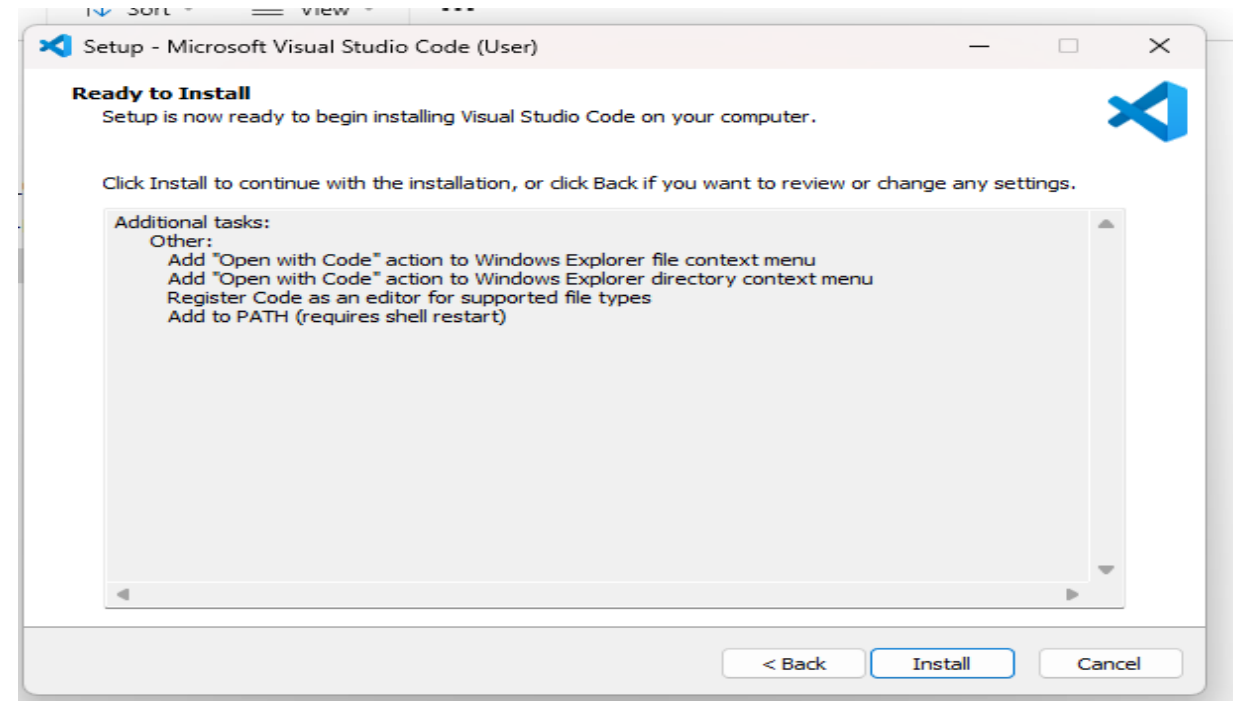
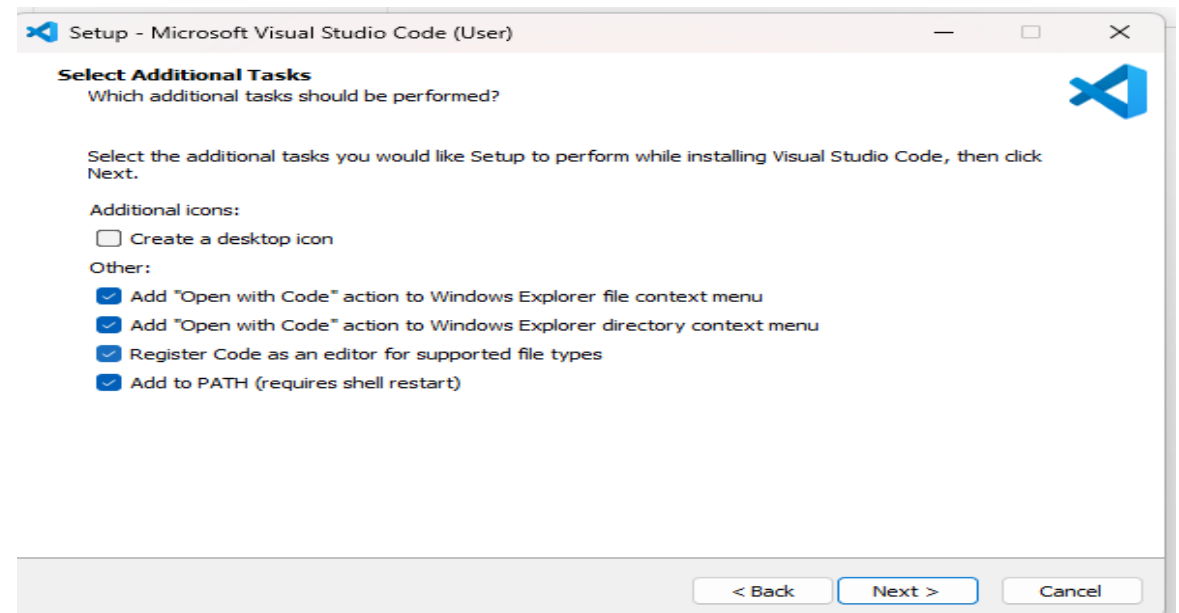
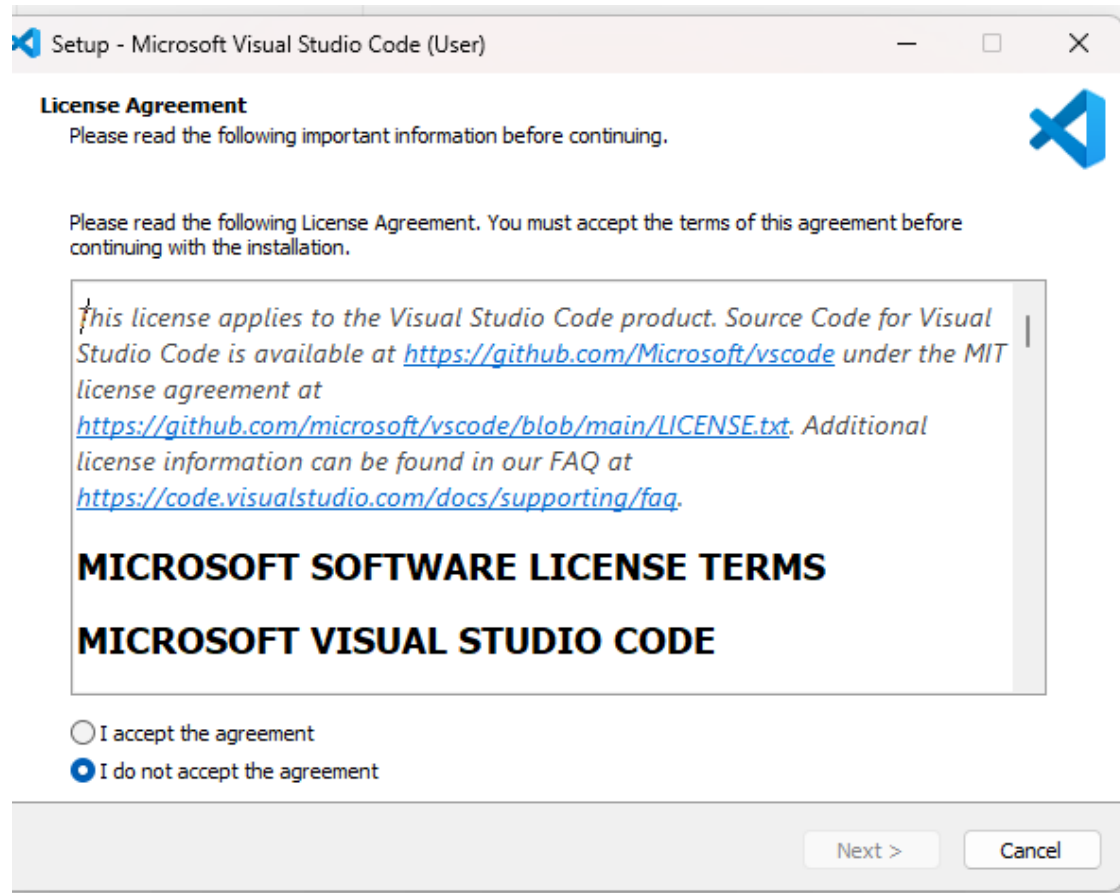


To set up a python environment for data science and machine learning, you typically need to install and configure the following:

1. Miniconda or Anaconda for managing python versions and creating isolated environments.
2. Git for version control and collaboration.
3. A code editor (VS Code) along with essential data science extensions
4. Core data science and machine learning packages (such as Numpy, Pandas, Matplotlib, Seaborn, Scikit-learn, PyTorch)
5. Configuration of Git and VS code to streamline workflow and development.



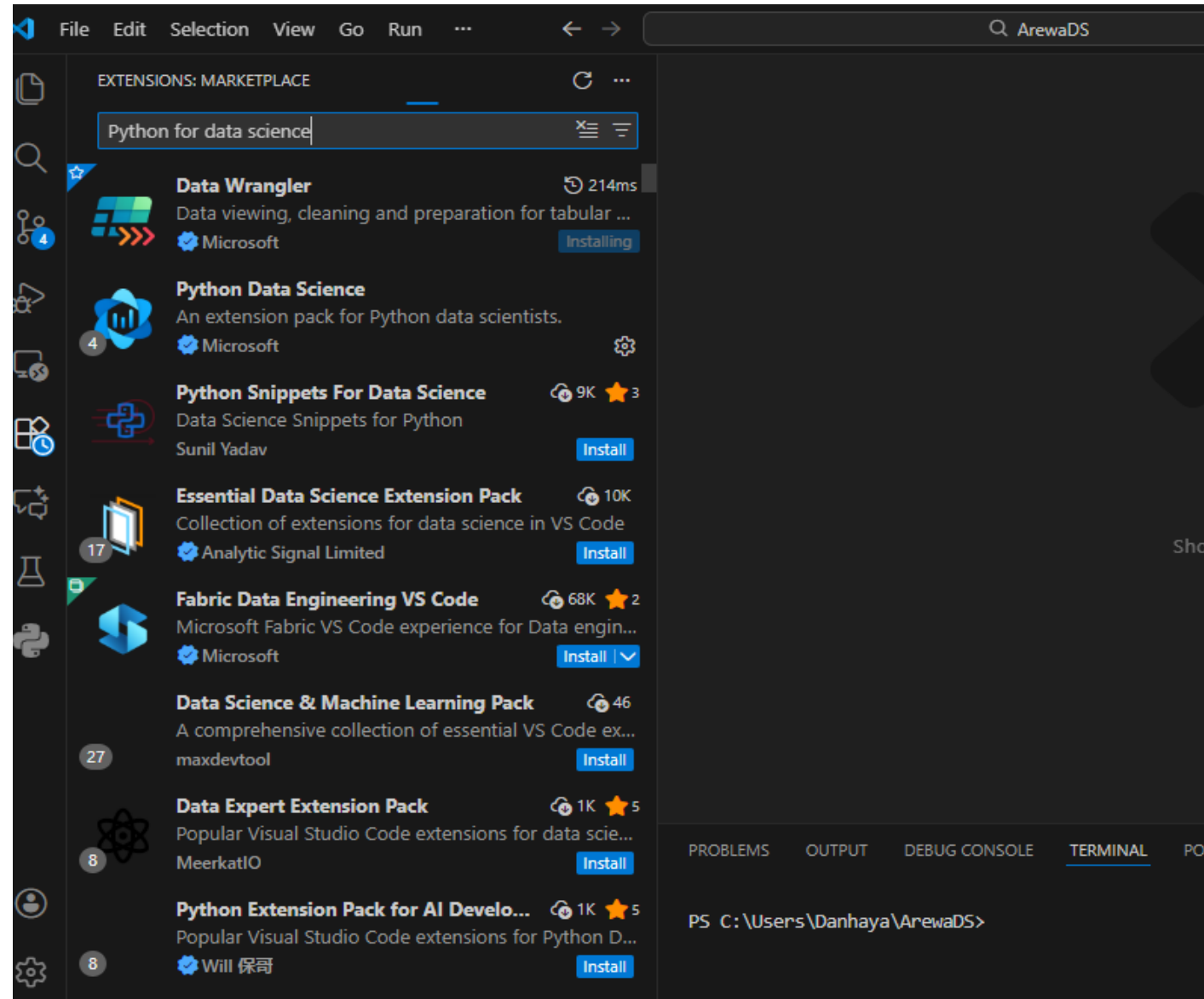
VS CODE



VS Code Extensions

You need to install the following extensions

- i. Python Data Science
- ii. Github copilot
- iii. Jupiter
- iv. Markdown all in one
- v. Python Environment Manager
- vi. Intellicode
- vii. VSCoDe-PDF



Core Data Science and Machine learning package

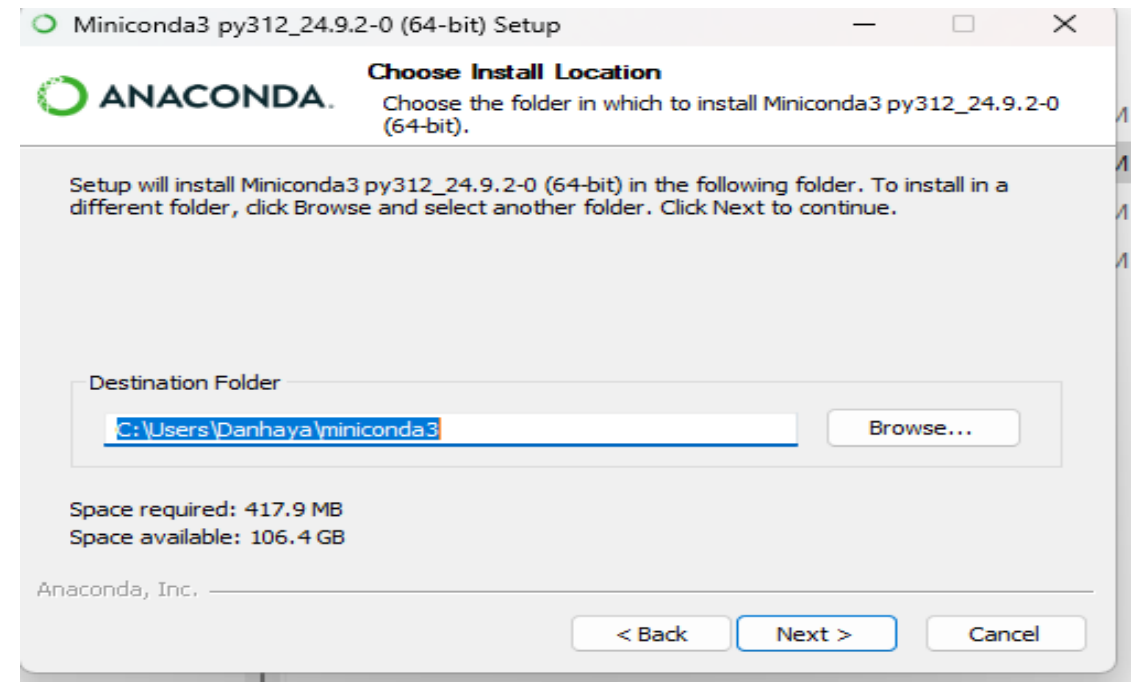
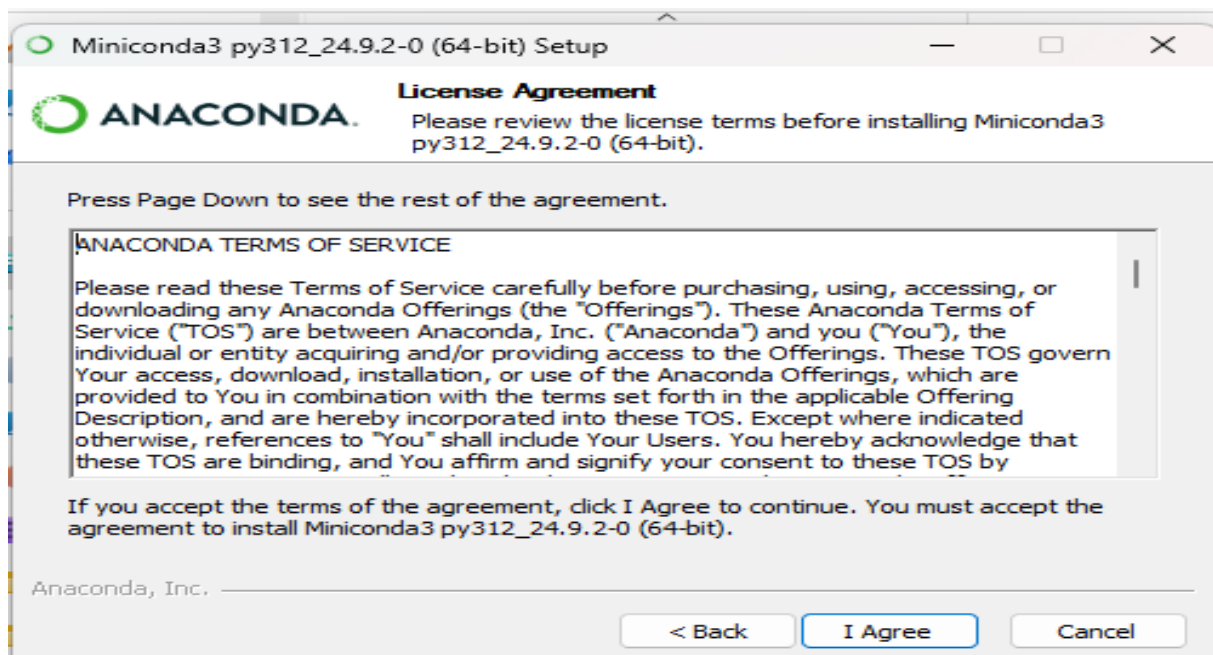
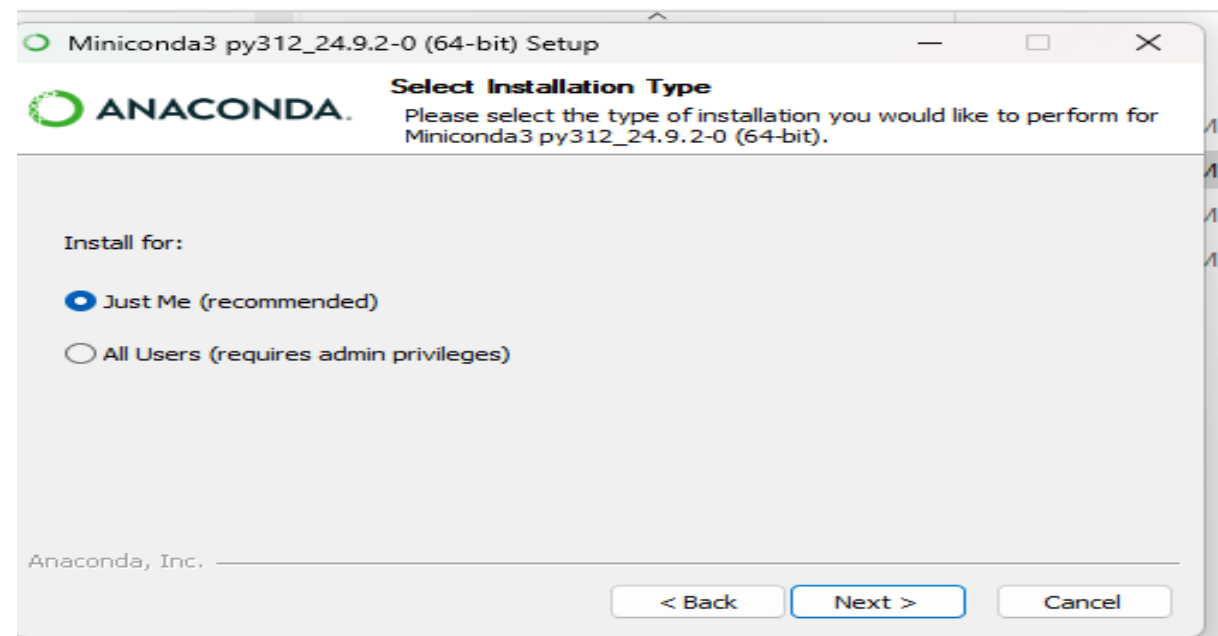
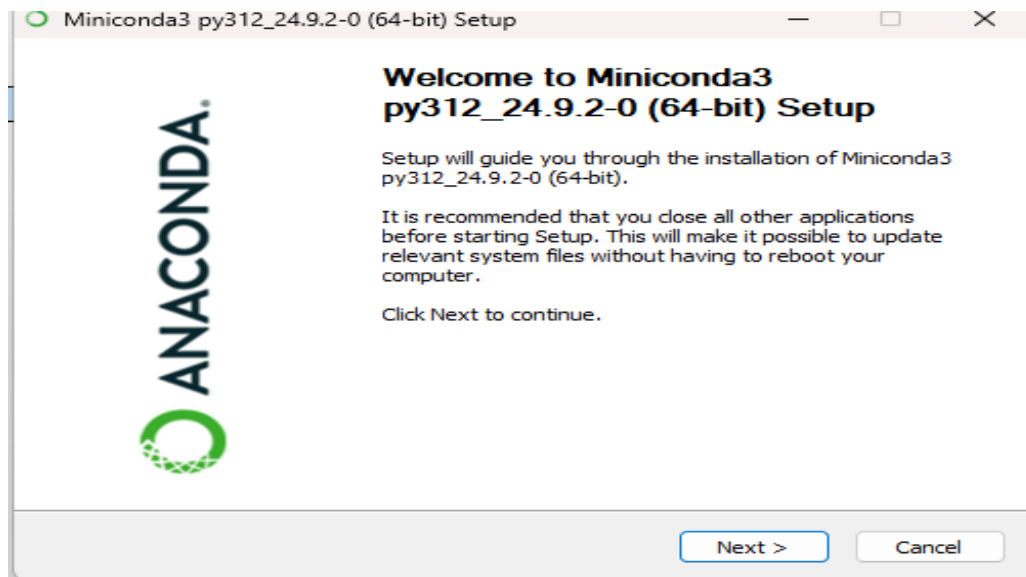
1. Numpy (Numerical Python)
2. Pandas
3. Scikit-Learn (Sklearn)
4. Matplotlib
5. Seaborn
6. PyTorch and many more

These packages can be install using pip install “package name” or conda install “package name” in the terminal e.g

- pip install pandas
- conda install matplotlib

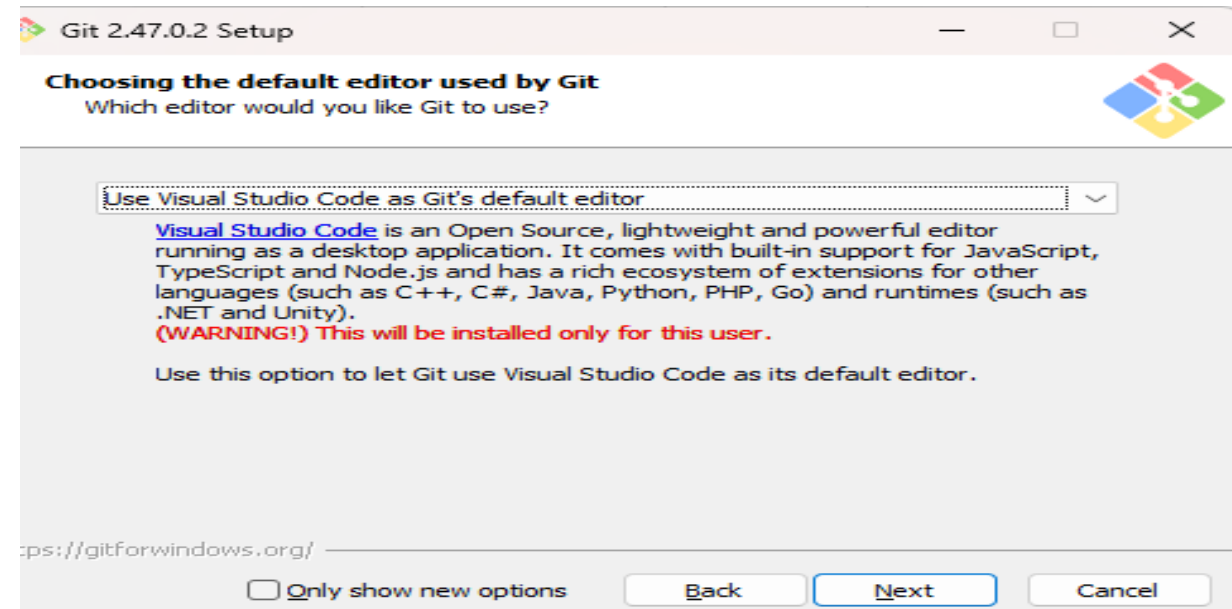
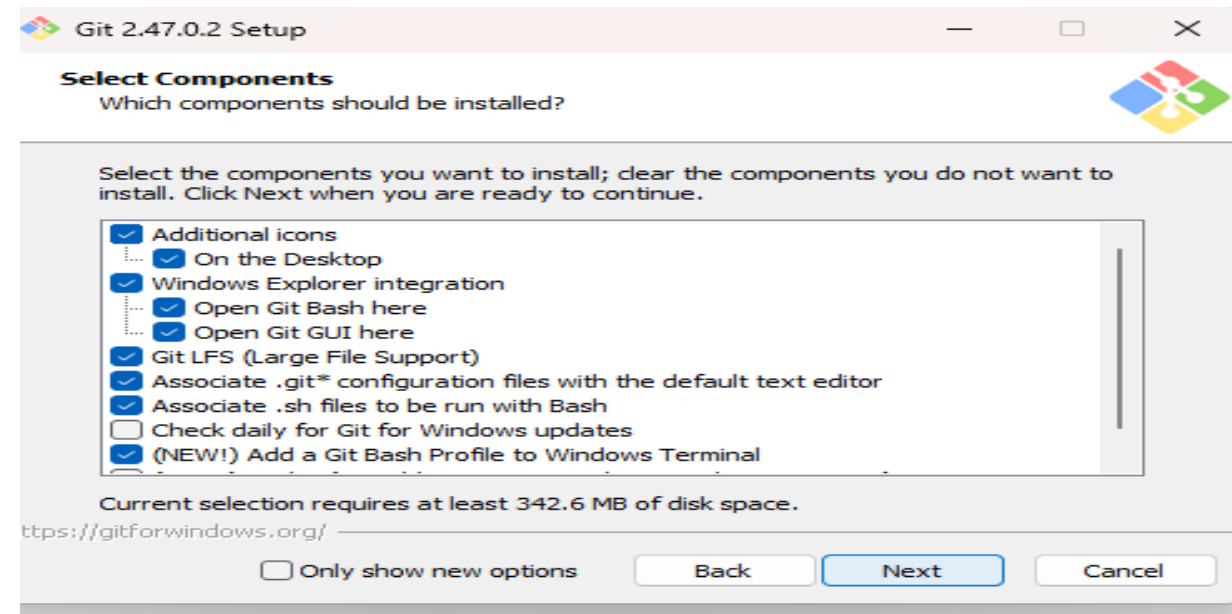


Miniconda



Follow the installation Wizard

- a. Select components
 - I. Keep everything default.
 - II. Make sure “open Git Bash here” and “Open Git GUI here” are checked
- b. Choose the default editor (VS Code recommended)
Use Visual Studio Code as Git’s default editor
- c. Let Git use the recommended PATH environment
Choose: “Git from the command line and also from 3rd –party software”
- d. Choose HTTPS
Select: “Use HTTPS instead of SSH”
- e. Choose default terminal emulator
Select: “Use MinTTY (the default terminal)”
- f. Other settings
 - a. Leave everything as default and continue clicking next.
 - b. Finally click install.



VERIFY GIT INSTALLATION

Open terminal and then type

➤ `git --version`

You should see something like

➤ `git version 2.xx.x` (your git was perfectly installed)

If you see error message something wrongs with your installation.

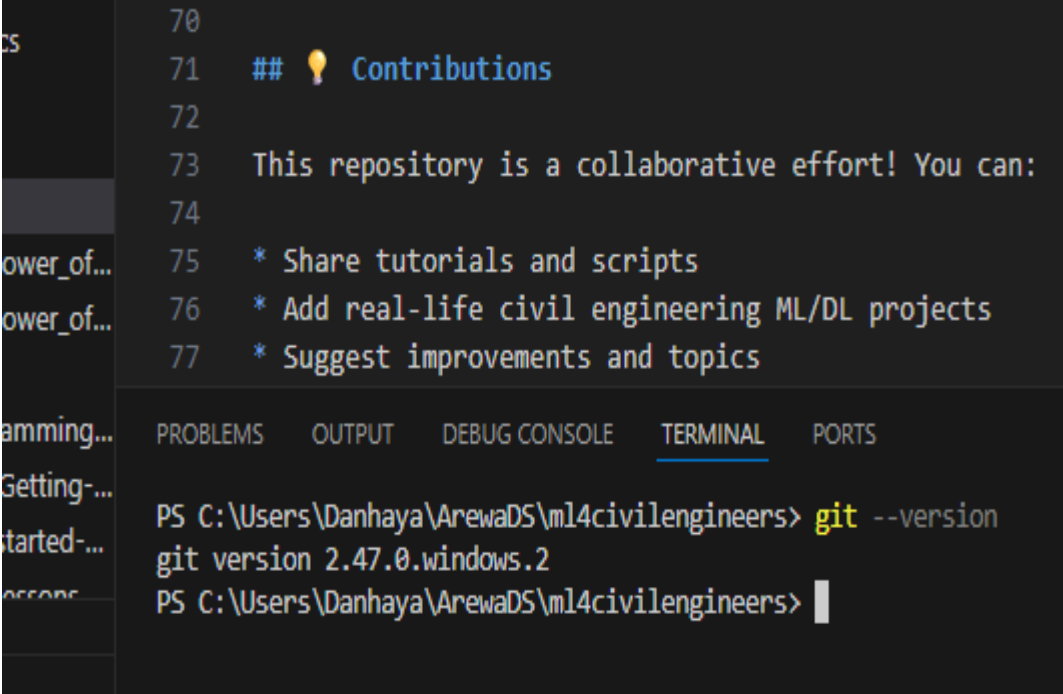
Why do we configure git?

Configuring git is important because git needs to know who is making changes in a project.

How do we configure git?

➤ `git config --global user.name "Your Username"`

➤ `git config --global user.email "someone@example.com"`



The image shows a screenshot of a Visual Studio Code editor with a terminal window open. The terminal displays the output of the command `git --version`. The output is `git version 2.47.0.windows.2`. Above the terminal, there is a file explorer view showing a directory structure. The terminal window has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, and TERMINAL (which is active). The terminal prompt is `PS C:\Users\Danhaya\ArewaDS\ml4civilengineers>`.

```
70
71 ## 💡 Contributions
72
73 This repository is a collaborative effort! You can:
74
75 * Share tutorials and scripts
76 * Add real-life civil engineering ML/DL projects
77 * Suggest improvements and topics
78
79 PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
80 PS C:\Users\Danhaya\ArewaDS\ml4civilengineers> git --version
81 git version 2.47.0.windows.2
82 PS C:\Users\Danhaya\ArewaDS\ml4civilengineers>
```

Thank you for coming this far
The best is yet to come!